

Heart Disease Prediction MLOps Project Report

End-to-End Machine Learning Solution & Production Pipeline

1. Title Page

Project Title: Heart Disease Prediction MLOps Project

BITS ID: 2024ad05313

Submitted By: Ankur Sharma

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2. Project Overview

This project demonstrates an end-to-end machine learning solution for predicting heart disease risk. The system combines data preprocessing, model training, a Flask-based web application, and deployment automation using Docker, GitHub Actions, and Azure.

The primary objective of this work is to move beyond notebook-based experimentation and implement a deployable, production-style MLOps workflow that can serve predictions through a user-friendly interface.

Key Deliverables

- A trained heart disease prediction model
- A Flask web application for single and batch predictions
- Automated CI/CD pipeline for testing and deployment
- Dockerized application for container-based deployment
- Azure deployment setup with health checks and web app hosting

3. EDA Findings

Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the heart disease dataset before model training.

Important Observations

- The dataset contains multiple medical attributes such as age, sex, cholesterol, blood pressure, and resting ECG results.
- Several features appear to be strongly correlated with heart disease risk.
- The target variable is categorical and suitable for classification models.
- Feature scaling and preprocessing are necessary to improve model stability.

EDA Summary

- Numerical attributes were analyzed for distribution and variance.
- Categorical variables were reviewed for their contribution to the target outcome.
- Missing values and inconsistent values were handled during preprocessing.

EDA helped in selecting appropriate preprocessing strategies and building a robust training pipeline.

4. Model Comparison

Several supervised classification algorithms were compared to identify the best-performing model for heart disease prediction.

Models Evaluated

- Logistic Regression
- Random Forest Classifier
- Support Vector Machine (SVM)

Comparison Criteria

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Result Summary

The best-performing model was selected based on the highest overall evaluation metrics and saved as a serialized artifact for prediction use.

The chosen model is stored in the artifacts directory and loaded during inference through the Flask prediction pipeline.

5. Architecture Diagram

[Architecture Flowchart Textual Representation]

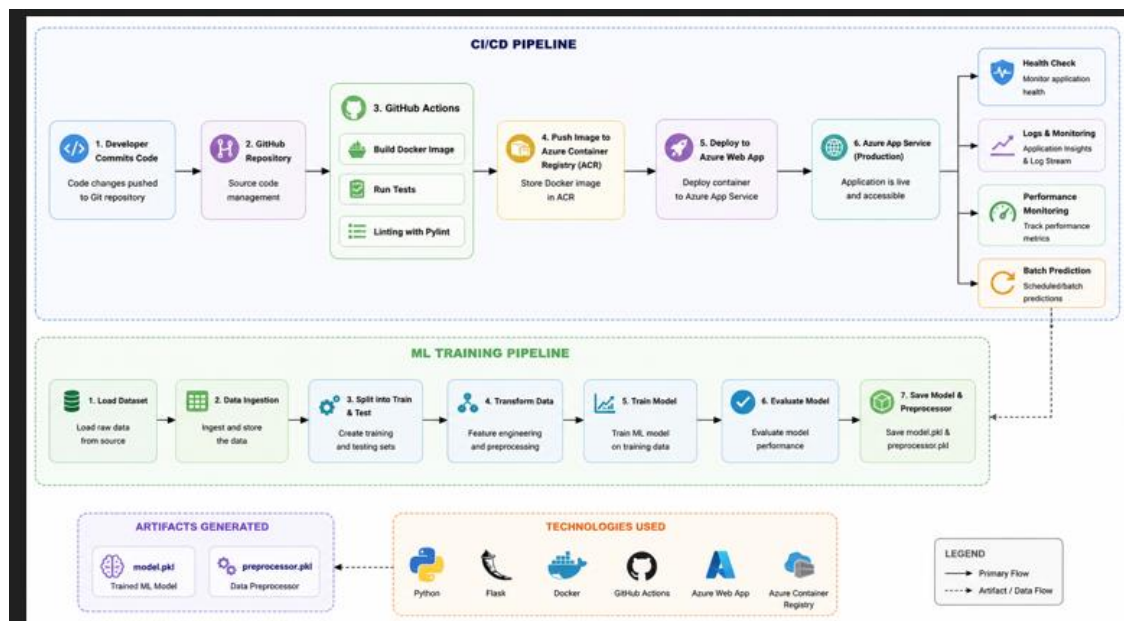
User -> Flask Web App -> Prediction API -> Trained ML Model -> Prediction Result

Training Pipeline:

Dataset -> Data Ingestion -> Data Transformation -> Model Training -> Model Evaluation
-> Artifacts (model.pkl)

DevOps Pipeline:

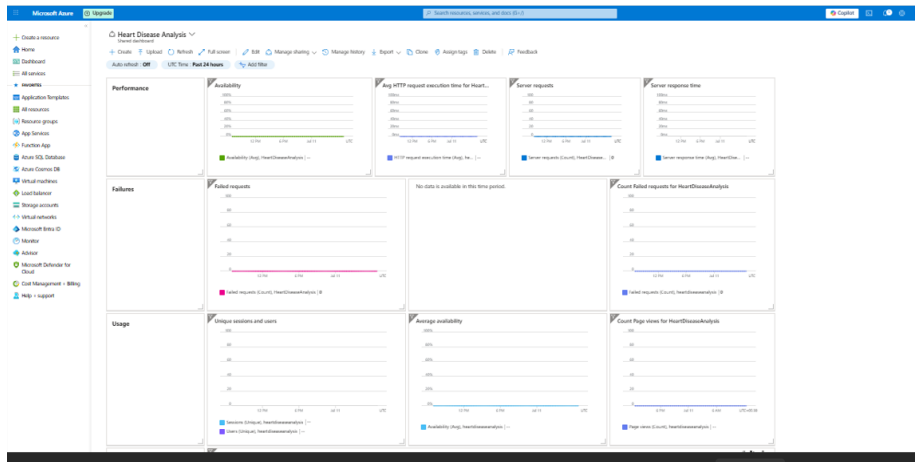
GitHub Repo -> GitHub Actions (flake8, pytest, Docker Build) -> Azure Container Registry -> Azure Web App



6. MLflow and Monitoring Screenshots

MLflow and monitoring tools were used to track experiments, model metrics, and system behavior during development.

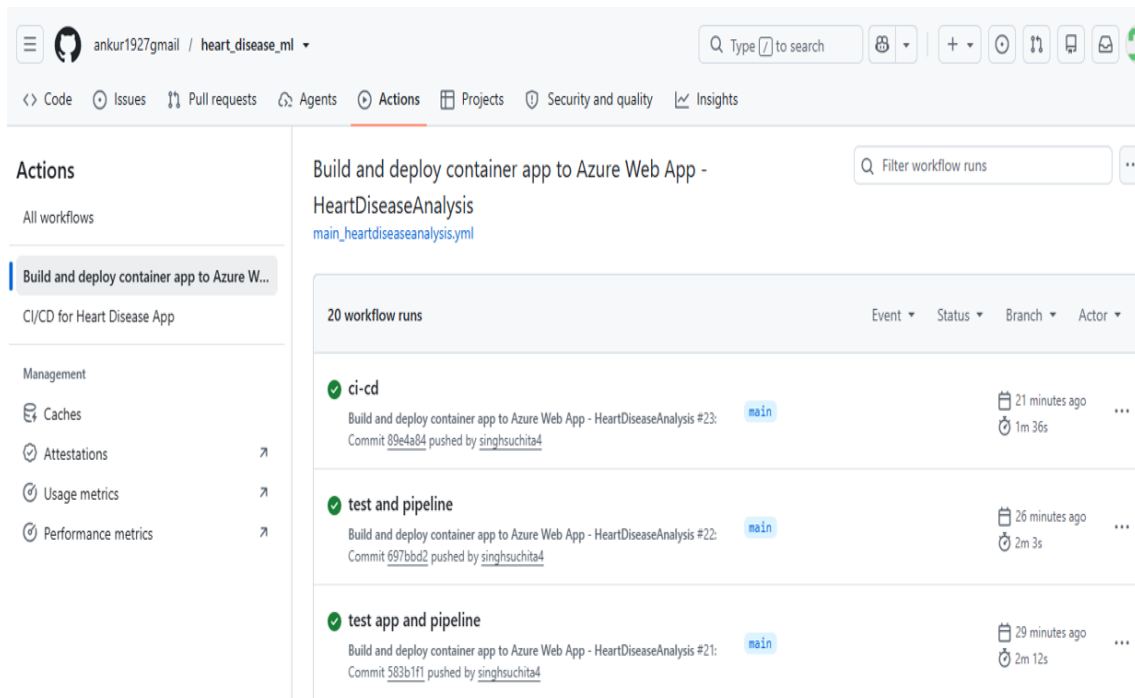
Screenshot 1: Monitoring Dashboard



7. CI/CD Screenshots

GitHub Actions was used to automate linting, testing, and deployment.

Screenshot 2: GitHub Actions Workflow



Screenshot 3: GitHub Repository Main View

 **heart_disease_ml** Public

Watch 0

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Add file

<> Code

 **ankur1927gmail** CICD Pipeline ✓

e776b8c · 10 hours ago 36 Commits

 .github/workflows	ci-cd	11 hours ago
 .vscode	EDA	last week
 artifacts	Model pickle file	5 days ago
 k8s	Deployment Manifest	11 hours ago
 notebook	data_transformation	last week
 screenshots	CICD Pipeline	10 hours ago
 src	changes linting	14 hours ago
 templates	workflow	yesterday
 tests	test and pipeline	11 hours ago
 .gitignore	Data Ingestion	last week
 Dockerfile	changes linting	14 hours ago
 README.md	screenshots	10 hours ago
 app.py	revoed promethieus references	12 hours ago
 requirements.txt	Deployment Manifest	11 hours ago
 setup.py	changes linting	14 hours ago

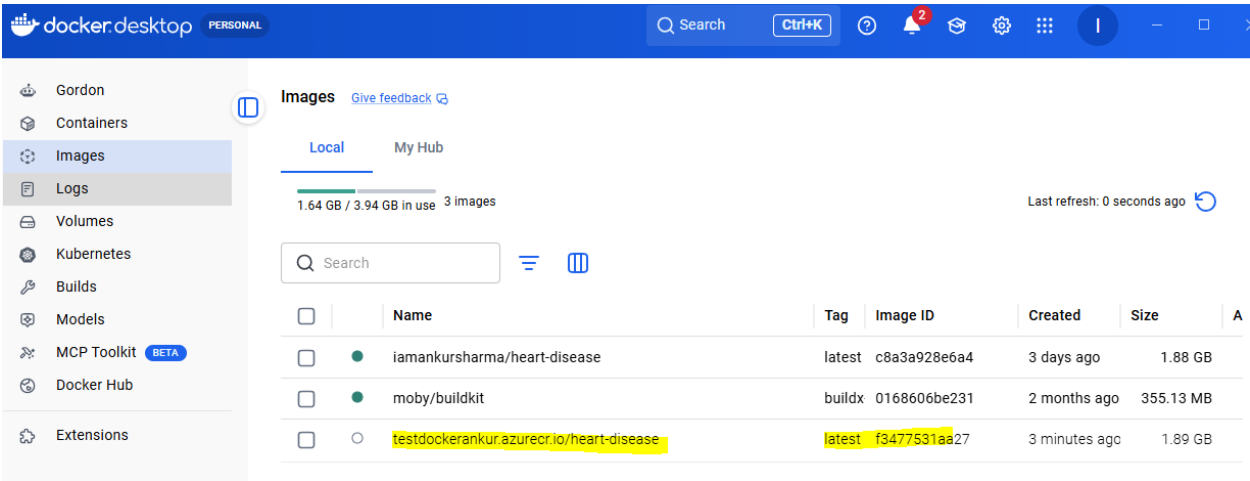
Screenshot 4: CI/CD Pipeline Summary



8. Deployment Screenshots

The application was containerized and deployed to Azure.

Screenshot 5: Azure Container Image



Screenshot 6: Application UI - Prediction Interface



Heart Disease Prediction System

Enter your medical data to predict heart disease risk

Manual Prediction

Bulk Prediction

Enter Your Medical Data

Age (years)

Range: 1 - 120 years

Sex

Select Gender

Chest Pain Type

Select Type

Resting Blood Pressure (mm Hg)

Typical: 90 - 180 mm Hg

Serum Cholesterol (mg/dl)

Typical: 125 - 200 mg/dl

Fasting Blood Sugar > 120 mg/dl

Select Value

Resting ECG Results

Select Result

Maximum Heart Rate Achieved

Typical: 60 - 200 bpm

Exercise Induced Angina

Select Value

ST Depression Induced by Exercise

Relative to rest (0 - 10)

ST Slope

Select Slope

Major Vessels Colored by Fluoroscopy

Select Number

Thalassemia (Thal)

Select Type

 PREDICT HEART DISEASE RISK

 CLEAR FORM



Prediction Result

Predicted Heart Disease Stage: 3

Screenshot 7: Application UI - Bulk Prediction View



Heart Disease Prediction System

Enter your medical data to predict heart disease risk

Manual Prediction

Bulk Predict JSON

Bulk Predict

Feature Reference

age — Age in years

sex — 0 = Female | 1 = Male

cp — Chest pain type: 1 = Typical Angina, 2 = Atypical Angina, 3 = Non-anginal, 4 = Asymptomatic

trestbps — Resting blood pressure (mm Hg)

chol — Serum cholesterol (mg/dl)

fbs — Fasting blood sugar > 120 mg/dl: 0 = No, 1 = Yes

restecg — Resting ECG: 0 = Normal, 1 = ST-T Abnormality, 2 = LV Hypertrophy

thalach — Maximum heart rate achieved

exang — Exercise induced angina: 0 = No, 1 = Yes

oldpeak — ST depression induced by exercise relative to rest

slope — ST slope: 1 = Upsloping, 2 = Flat, 3 = Downsloping

ca — Number of major vessels (0-3) colored by fluoroscopy

thal — Thalassemia: 3 = Normal, 6 = Fixed Defect, 7 = Reversible Defect

Num/Heart disease stage — 0 = No Heart Disease, 1 = Heart Disease Stage 1, 2 = Heart Disease Stage 2, 3 = Heart Disease Stage 3, 4 = Heart Disease Stage 4

Upload JSON file

Choose File

No file chosen

Only JSON files are allowed, maximum size 200MB.

PREDICT FROM JSON

CLEAR RESULTS

Screenshot 8: Project Structure Reference

```
heart_disease_mlops/
├── .github/
│   └── workflows/
│       ├── ci-cd.yml
│       └── main_heartdiseaseanalysis.yml
├── app.py
├── Dockerfile
├── requirements.txt
├── setup.py
├── README.md
├── artifacts/
│   ├── data.csv
│   ├── test.csv
│   ├── train.csv
│   ├── model.pkl
│   └── preprocessor.pkl
├── k8s/
│   ├── deployment.yaml
│   ├── ingress.yaml
│   └── hpa.yaml
├── logs/
├── notebook/
│   ├── EDA_HEART_DISEASE.ipynb
│   ├── MODEL_TRAINING.ipynb
│   └── data/
│       ├── heart_disease_uci.csv
│       └── preprocessed.csv
├── screenshots/
├── src/
│   ├── __init__.py
│   ├── exception.py
│   ├── logger.py
│   ├── utils.py
│   ├── components/
│   │   ├── __init__.py
│   │   ├── data_ingestion.py
│   │   ├── data_transformation.py
│   │   └── model_trainer.py
│   └── pipeline/
│       ├── __init__.py
│       ├── predict_pipeline.py
│       └── train_pipeline.py
├── templates/
│   ├── home.html
│   └── index.html
├── tests/
│   └── test_app_and_pipeline.py
└── venv312/
```

These deployment and UI screenshots provide visual evidence of the completed solution and its hosted application experience.

9. Setup Instructions

Prerequisites

- Python 3.11
- Docker
- Git
- Azure account (for deployment)

Clone the Repository

```
git clone <repository-url>  
cd heart_disease_mlops
```

Create a Virtual Environment

```
python -m venv venv  
source venv/bin/activate
```

Install Dependencies

```
pip install -r requirements.txt
```

Run the Application Locally

```
python app.py
```

The application will run at:

```
http://localhost:8000
```

Run Tests

```
pytest -q
```

Run Lint Checks

```
python -m flake8 . --exclude=.git,venv,venv312,.venv,__pycache__
```

Build Docker Image

```
docker build -t heart-disease .
```

10. Repository Link

Repository: https://github.com/ankur1927gmail/heart_disease_ml

API Link: <https://heartdiseaseanalysis-c6bbb8fgbdhdajby.centralus-01.azurewebsites.net>

11. Conclusion

This project successfully demonstrates an end-to-end MLOps workflow for heart disease prediction. It covers data analysis, model training, deployment, automation, and cloud delivery in a structured and practical manner.

The solution provides a strong foundation for future enhancement such as monitoring, retraining, model versioning, and production-grade observability.